

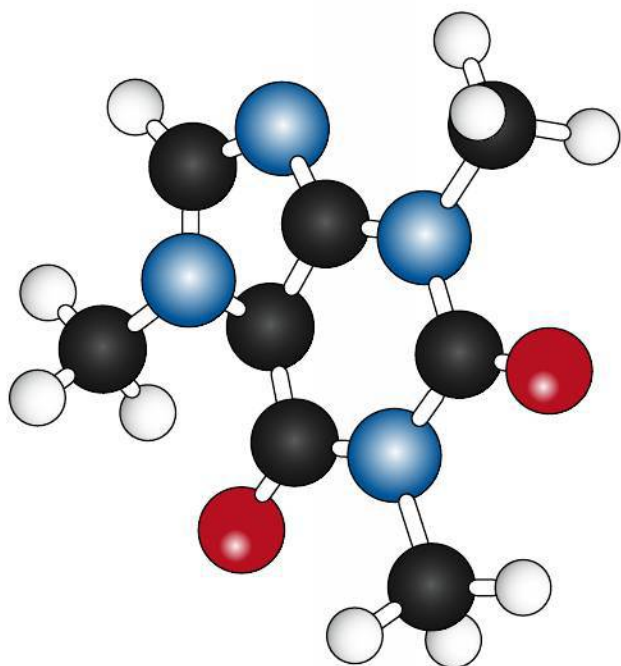


Figure 1.10 A molecular model of the stimulant, caffeine. The atoms are colour coded as follows: black = carbon, grey = hydrogen, red = oxygen, blue = nitrogen. Can you deduce the molecular formula, the empirical formula, and the molar mass of caffeine?

Solution

First calculate the mass of the empirical formula:

$$\text{mass}(\text{HgCl}) = 200.59 + 35.45 = 236.04 \text{ g mol}^{-1}$$

$$(236.04) \times x = M = 472.08$$

$$\therefore x = 2$$

$$\text{molecular formula} = \text{Hg}_2\text{Cl}_2$$

Combustion analysis usually gives data on the mass of compounds formed

The data presented so far may suggest that combustion analysis directly gives information on the relative masses of individual elements in a compound. In fact this is rarely the case, but instead elements are converted into new compounds, typically their oxides, by reaction with oxygen. So the primary data obtained are the masses of carbon dioxide, water, sulfur dioxide, and so on, which are measured by infra-red absorption, as described in Chapter 11. Processing these data simply involves an extra step.

Worked example

A 0.5438 g sample of a compound known to contain only carbon, hydrogen, and oxygen was burned completely in oxygen. The products were 1.0390 g CO_2 and 0.6369 g H_2O . Determine the empirical formula of the compound.

Solution

First we must convert the mass of each compound to moles in the usual way. From the number of moles of CO_2 and H_2O we can deduce the number of moles of C atoms and H atoms.

$$n(\text{CO}_2) = \frac{1.0390 \text{ g}}{12.01 + (16.00 \times 2) \text{ g mol}^{-1}} = 0.02361 \text{ mol CO}_2 \Rightarrow 0.02361 \text{ mol C atoms}$$

$$n(\text{H}_2\text{O}) = \frac{0.6369 \text{ g}}{(1.01 \times 2) + 16.00 \text{ g mol}^{-1}} = 0.03534 \text{ mol H}_2\text{O} \Rightarrow 0.03534 \times 2 = 0.07068 \text{ mol H atoms}$$

In order to know the mass of O in the original sample, we must convert the number of moles of C and H atoms to mass by multiplying by their molar mass, M.

$$\text{mass C} = 0.02361 \text{ mol} \times 12.01 \text{ g mol}^{-1} = 0.2836 \text{ g}$$

$$\text{mass H} = 0.07068 \text{ mol} \times 1.01 \text{ g mol}^{-1} = 0.07139 \text{ g}$$

$$\therefore \text{mass O} = 0.5438 - (0.2836 + 0.07139) = 0.1888 \text{ g}$$

$$\text{mol O atoms} = \frac{0.1888 \text{ g}}{16.00 \text{ g mol}^{-1}} = 0.01180 \text{ mol}$$

Now we can proceed as with the previous examples, converting mass of O to moles and then comparing the mole ratios.

	Carbon, C	Hydrogen, H	Oxygen, O
mass / g			0.1888
moles	0.02361	0.07068	0.01180
divide by smallest	2.001	5.999	1.000
nearest whole number ratio	2	6	1

So the empirical formula is $\text{C}_2\text{H}_6\text{O}$.

Exercises

24 Give the empirical formulas of the following compounds:

- (a) ethyne, C_2H_2
- (b) glucose, $\text{C}_6\text{H}_{12}\text{O}_6$
- (c) sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
- (d) octane, C_8H_{18}
- (e) oct-1-yne, C_8H_{14}
- (f) ethanoic acid, CH_3COOH

25 A sample of a compound contains only the elements sodium, sulfur, and oxygen. It is found by analysis to contain 0.979 g Na, 1.365 g S, and 1.021 g O. Determine its empirical formula.

26 A sample of a hydrated compound was analysed and found to contain 2.10 g Co, 1.14 g S, 2.28 g O, and 4.50 g H_2O . Determine its empirical formula.

27 A street drug has the following composition: 83.89% C, 10.35% H, 5.76% N. Determine its empirical formula.

28 The following compounds are used in the production of fertilizers. Determine which has the highest percentage by mass of nitrogen: NH_3 , $\text{CO}(\text{NH}_2)_2$, $(\text{NH}_4)_2\text{SO}_4$.

29 A compound has a formula M_3N where M is a metal element and N is nitrogen. It contains 0.673 g of N per gram of the metal M. Determine the relative atomic mass of M and so its identity.

30 Compounds of cadmium are used in the construction of photocells. Deduce which of the following has the highest percentage by mass of cadmium: CdS, CdSe, CdTe.

31 Benzene is a hydrocarbon, a compound of carbon and hydrogen only. It is found to contain 7.74% H by mass. Its molar mass is 78.10 g mol^{-1} . Determine its empirical and molecular formulas.

32 A weak acid has a molar mass of 162 g mol^{-1} . Analysis of a 0.8821 g sample showed the composition by mass is 0.0220 g H, 0.3374 g P, and the remainder was O. Determine its empirical and molecular formulas.

33 ATP is an important molecule in living cells. A sample with a mass of 0.8138 g was analysed and found to contain 0.1927 g C, 0.02590 g H, 0.1124 g N, and 0.1491 g P. The remainder was O. Determine the empirical formula of ATP. Its formula mass was found to be 507 g mol^{-1} . Determine its molecular formula.

34 A 0.30 g sample of a compound that contains only carbon, hydrogen, and oxygen was burned in excess oxygen. The products were 0.66 g of carbon dioxide and 0.36 g of water. Determine the empirical formula of the compound.

35 You are asked to write your name on a suitable surface, using a piece of chalk that is pure calcium carbonate, CaCO_3 . How could you calculate the number of carbon atoms in your signature?



How many atoms of calcium are in Avogadro's name written in chalk?

1.3 Reacting masses and volumes

Understandings:

- Reactants can be either limiting or excess.
- The experimental yield can be different from the theoretical yield.
- Avogadro's law enables the mole ratio of reacting gases to be determined from volumes of the gases.
- The molar volume of an ideal gas is a constant at specified temperature and pressure.

Guidance

Values for the molar volume of an ideal gas are given in the IB data booklet in Section 2.

- The molar concentration of a solution is determined by the amount of solute and the volume of solution.

Guidance

The use of square brackets to denote molar concentration is required.

- A standard solution is one of known concentration.

Applications and skills:

- Solution of problems relating to reacting quantities, limiting and excess reactants, and theoretical, experimental, and percentage yields.
- Calculation of reacting volumes of gases using Avogadro's law.
- Solution of problems and analysis of graphs involving the relationship between temperature, pressure, and volume for a fixed mass of an ideal gas.
- Solution of problems relating to the ideal gas equation.

Guidance

The ideal gas equation, $PV = nRT$, and the value of the gas constant (R), are given in the IB data booklet in Sections 1 and 2.

- Explanation of the deviation of real gases from ideal behaviour at low temperature and high pressure.
- Obtaining and using experimental values to calculate the molar mass of a gas from the ideal gas equation.
- Solution of problems involving molar concentration, amount of solute, and volume of solution.

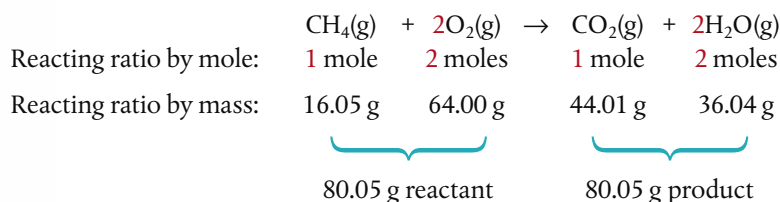
Guidance

Units of concentration to include: g dm^{-3} , mol dm^{-3} , and parts per million (ppm).

- Use of the experimental method of titration to calculate the concentration of a solution by reference to a standard solution.

Chemical equations show reactants combining in a fixed molar ratio

Chemical change, as summarized in chemical equations, is simply an expression of reactants combining in fixed ratios to form products. The most convenient means to express this ratio is as moles, as that gives us a means of relating the number of particles that react to the mass that we can measure. So, for example, when methane, CH_4 , burns in air, we can conclude the following, all from the balanced chemical equation:



(The figures for total mass of reactant and product are just a check, as we know something would be wrong if they did not equate.)



The Bunsen burner controls the combustion of methane, CH_4 , and is often used as a source of heat in the laboratory.

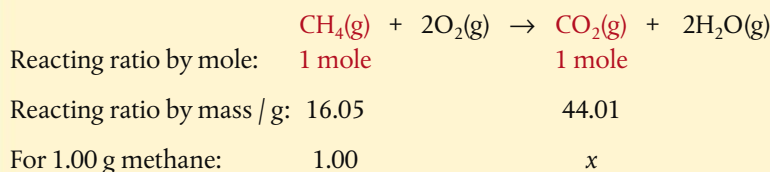
This simple interpretation of equations, going directly from coefficients to molar ratios, opens the door to a wide range of calculations involving reacting masses.

Worked example

Calculate the mass of carbon dioxide produced from the complete combustion of 1.00 g of methane.

Solution

Write the balanced equation and deduce the mole ratio as above. Then pick out from the question the terms that we need to analyse, here they are marked in red; these are the species where we need to convert moles to grams.



Now solve the ratio, shown here using cross-multiplication, to determine the value of x.

$$\frac{\text{g CH}_4}{\text{g CO}_2} = \frac{16.05}{44.01} = \frac{1.00}{x}$$

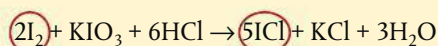
$$x = \frac{1.00 \times 44.01}{16.05} = 2.74 \text{ g CO}_2$$

All questions on reacting ratios involve a variation of this approach:

- write the balanced equation;
- work out the mole ratio for the species identified in the question;
- work out the reacting ratio by mass for these species, using $m = nM$;
- insert the data from the question and solve the ratio.

Worked example

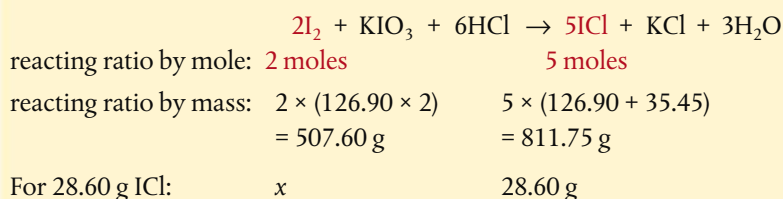
Iodine chloride, ICl, can be made by the following reaction:



Calculate the mass of iodine, I₂, needed to prepare 28.60 g of ICl by this reaction.

Solution

The relevant terms from the question are I₂ and ICl, so these are our focus.



$$\frac{\text{g I}_2}{\text{g ICl}} = \frac{507.60}{811.75} = \frac{x}{28.60}$$

$$x = \frac{507.60 \times 28.60}{811.75} \text{ g I}_2 = 17.88 \text{ g I}_2$$



The term **carbon footprint** refers to the mass of carbon dioxide and other greenhouse gases such as methane that an individual emits in a 1-year period. It is often expressed as the carbon dioxide equivalent, or CO₂e, to represent the total climate change impact of all the greenhouse gases caused by an item or activity. It includes emissions from fuels used in transport, services such as heating, production and consumption of food, and the direct and indirect emissions from manufactured goods and construction. It is extremely difficult to measure all sources accurately, but the fundamental concept uses the type of calculation shown here. The carbon footprint is a measure of an individual's consumption of resources, and suggests the link between this and the enhanced greenhouse effect. There is further discussion of carbon footprints in Chapter 12.



Remember to convert between mass and moles by using $n = \frac{m}{M}$ or $m = nM$



It is common for people to waste time solving questions like this by doing the mole to gram conversions for *all* the species represented in the equation. You can save yourself a lot of trouble by focusing *only* on the terms indicated in the question, as shown in this example.

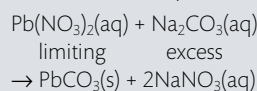
The theoretical yield is determined by the limiting reactant

Imagine that you are following a recipe to make 12 cookies. It calls for you to mix two eggs with four cups of flour. The problem is that you only have one egg. You will quickly realize that this means you can use only two cups of flour and end up with only six cookies. We could say that the number of eggs *limited* the amount of product.

In many chemical reactions the relative amounts of reactants available to react together will similarly affect the amount of product. The reactant that determines the quantity of product is known as the **limiting reactant**. Other reactants will therefore not be fully used, and are said to be in **excess**. Identifying the limiting reactant is therefore a crucial step before we can calculate the expected quantity of product. The **theoretical yield**, which is usually expressed in grams or moles, refers to the maximum amount of product obtainable, assuming 100% of the limiting reactant is converted to product.

Note that identification of the limiting reactant depends on the *mole ratios* in the balanced chemical equation for the reaction. This means that if reactant quantities are given in grams, they must first be converted to moles.

The concept of limiting reactant is often useful in the design of experiments and synthetic processes. By deliberately making one reactant available in an amount greater than that determined by its mole ratio in the balanced equation, it ensures that the other reactant is limiting and so will be fully used up. For example, in order to remove lead ions in lead nitrate from a contaminated water supply, sodium carbonate, Na_2CO_3 , is added to precipitate lead carbonate, as shown in the equation:

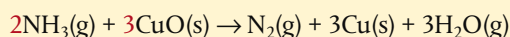


By using excess Na_2CO_3 , this ensures that *all* the lead ions react and so are removed from the water supply.



Worked example

Nitrogen gas (N_2) can be prepared from this reaction:



If 18.1 g NH_3 are reacted with 90.40 g CuO , determine the mass of N_2 that can be formed.

Solution

First we must determine the limiting reactant. We convert the mass of reactants to moles, and then compare the mole ratio in the balanced equation with the mole ratio of reactants given.

$$n(\text{NH}_3) = \frac{18.1 \text{ g}}{14.01 + (3 \times 1.01) \text{ g mol}^{-1}} = 1.06 \text{ mol NH}_3$$

$$n(\text{CuO}) = \frac{90.40 \text{ g}}{63.55 + 16.00 \text{ g mol}^{-1}} = 1.14 \text{ mol CuO}$$

$$\text{mole ratio from equation: } \frac{\text{NH}_3}{\text{CuO}} = \frac{2}{3} = 0.667$$

$$\text{mole ratio from given masses: } \frac{\text{NH}_3}{\text{CuO}} = \frac{1.06}{1.14} = 0.930$$

As the ratio $\text{NH}_3 : \text{CuO}$ of the given masses is *larger* than the required ratio in the equation, it means NH_3 is in excess and CuO is the limiting reactant.

This means that the amount of N_2 that can form will be determined by the amount of CuO . This is now similar to the earlier questions, where we write out the equation and focus on the terms identified in the question.



reacting ratio by mole: 3 moles 1 mole

for 1.14 moles CuO : 1.14 x

$$\text{mole ratio } \frac{\text{CuO}}{\text{N}_2} = \frac{3}{1} = \frac{1.14}{x} \quad \therefore x = \frac{1.14 \times 1}{3} = 0.380 \text{ mol N}_2$$

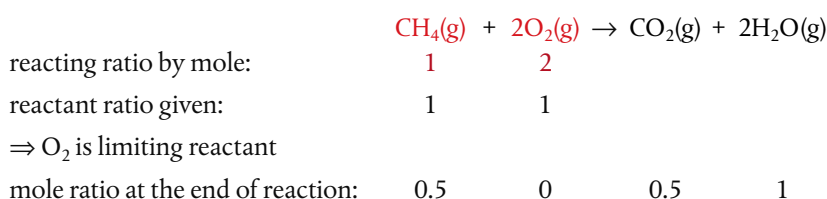
$$\therefore \text{mass N}_2 = 0.380 \text{ mol} \times M(\text{N}_2) \text{ g mol}^{-1} = 0.380 \text{ mol} \times 28.02 \text{ g mol}^{-1} = 10.7 \text{ g N}_2$$

There are alternate approaches to determining the limiting reactant, such as calculating which given amount of reactant would yield the smallest amount of product. But in essence, all questions on limiting reactant and theoretical yield involve comparing the mole ratio of given masses of reactants with the coefficients in the equation. This is a summary of the steps:

- write the balanced equation and focus on the mole ratio of reactants;
- convert the given mass of reactants to moles;
- compare the given mole ratios with the ratio of coefficients in the equation;
- identify the limiting reactant from the above ratios;
- calculate the moles of product from the given moles of limiting reactant.

Sometimes it is useful to measure how much excess reactant will remain when all the limiting reactant has been used up and the reaction stops. One example of this is a technique called **back-titration**, which analyses excess acid or alkali after a reaction is complete, and so indirectly measures the amount of a limiting reactant. This is explained on page 50.

A simple example of how to calculate the excess is shown below, using an example of burning CH_4 when 1 mole of CH_4 and 1 mole of O_2 are supplied.



So 0.5 moles CH_4 will be un-reacted at the end of the reaction.

We will study more examples of this type of question in the next section.

The percentage yield can be calculated from the experimental and theoretical yields

The answer to the Worked example above, 10.7 g N_2 is known as the theoretical yield, because it assumes that *all* of the CuO was converted to N_2 with no loss, impurities present, wastage, or incomplete reaction. In reality, all of the above happen to different extents in most chemical reactions, and so the theoretical yield is usually different from the actual or **experimental yield**.

When we compare the experimental yield with the theoretical yield, we get a measure of the efficiency of the conversion of reactants to products. This is usually expressed as the **percentage yield**, defined as follows:

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

In your own experiments, you may often be able to calculate the percentage yield of product in evaluating the results. In industry, this is a very important calculation to determine the efficiency of a process such as the synthesis of a drug in the pharmaceutical industry. Many aspects of green chemistry focus on ways to increase the yield of product by reducing wastage.

The limiting reactant determines the amount of product that can form. The theoretical yield is the quantity of product that can form from the complete conversion of the limiting reactant.



Industrial plants such as this oil refinery need to be able to track the efficiency of the chemical reactions taking place. Measuring the yield of product is an essential part of this.

CHALLENGE YOURSELF

- 4 Percentage yield and the atom economy are different concepts, but both can be used to assess aspects of the overall efficiency of a chemical process. See if you can find a reaction that has a high percentage yield under certain conditions, but a low atom economy.

Worked example

The previous Worked example on the synthesis of N_2 from NH_3 and CuO had a theoretical yield of 10.7 g N_2 from the starting amounts of reactants. Under the same conditions, an experiment produced 8.35 g N_2 . Determine the percentage yield.

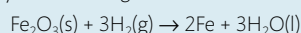
Solution

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

$$\therefore \text{percentage yield} = \frac{8.35 \text{ g}}{10.7 \text{ g}} \times 100 = 78.0\%$$

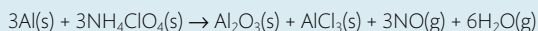
Exercises

- 36 Iron ore can be reduced to iron by the following reaction:



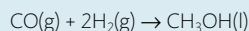
- How many moles of Fe can be made from 1.25 moles of Fe_2O_3 ?
 - How many moles of H_2 are needed to make 3.75 moles of Fe?
 - If the reaction yields 12.50 moles of H_2O , what mass of Fe_2O_3 was used up?
- 37 Lighters commonly use butane, C_4H_{10} , as the fuel.
- Formulate the equation for the combustion of butane.
 - Determine the mass of butane that burned when 2.46 g water were produced.

- 38 Booster rockets for the space shuttle use the following reaction:



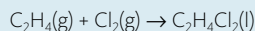
Calculate the mass of NH_4ClO_4 that should be added to this fuel mixture to react completely with every kilogram of Al.

- 39 Limestone is mostly calcium carbonate, CaCO_3 , but also contains other minerals. When heated, the CaCO_3 decomposes into CaO and CO_2 . A 1.605 g sample of limestone was heated and gave off 0.657 g of CO_2 .
- Formulate the equation for the thermal decomposition of calcium carbonate.
 - Determine the percentage mass of CaCO_3 in the limestone.
 - State the assumptions that you are making in this calculation.
- 40 Methanol, CH_3OH , is a useful fuel that can be made as follows:



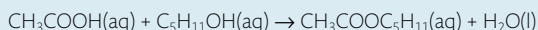
A reaction mixture used 12.0 g of H_2 and 74.5 g of CO .

- Determine the theoretical yield of CH_3OH .
 - Calculate the amount of the excess reactant that remains unchanged at the end of the reaction.
- 41 The dry-cleaning solvent 1,2-dichloroethane, $\text{C}_2\text{H}_4\text{Cl}_2$, is prepared from the following reaction:



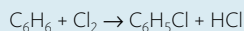
Determine the mass of product that can be formed from 15.40 g of C_2H_4 and 3.74 g of Cl_2 .

- 42 Calcium carbonate, CaCO_3 , is able to remove sulfur dioxide, SO_2 , from waste gases by a reaction in which they react in a 1:1 stoichiometric ratio to form equimolar amounts of CaSO_3 . When 255 g of CaCO_3 reacted with 135 g of SO_2 , 198 g of CaSO_3 were formed. Determine the percentage yield of CaSO_3 .
- 43 Pentyl ethanoate, $\text{CH}_3\text{COOC}_5\text{H}_{11}$, which smells like bananas, is produced from the esterification reaction:



A reaction uses 3.58 g of CH_3COOH and 4.75 g of $\text{C}_5\text{H}_{11}\text{OH}$ and has a yield of 45.00%. Determine the mass of ester that forms.

- 44 A chemist has to make a 100 g sample of chlorobenzene, $\text{C}_6\text{H}_5\text{Cl}$, from the following reaction:



Determine the minimum quantity of benzene, C_6H_6 , that can be used to achieve this with a yield of 65%.

Avogadro's law directly relates gas volumes to moles

All the examples above use mass as a way to measure amount, the number of moles. But in the laboratory we often work with liquids and gases, where *volume* is a more convenient measure. So what is the relationship between gas volume and number of moles?

Consider the following demonstration, where two gas jars are each filled with different gases – hydrogen (H_2) in flask A and bromine (Br_2) in flask B. The flasks are at the same temperature and pressure and have equal volumes.

It is known, from many experimental measurements on gas volumes, that the number of particles in the two flasks above is the same. At first this might seem surprising – after all bromine molecules are much larger and heavier than hydrogen molecules. But we need to consider the nature of the gaseous state, and as we learned on page 9, remember that the particles in a gas are widely spaced out with only negligible forces between them. In simple terms, most of a gas volume is empty space. And for this reason the chemical nature of the gas is irrelevant to its volume. Gas volume is determined only by the number of particles and by the temperature and pressure.

This understanding is known as Avogadro's law, which states that:

Equal volumes of all gases, when measured at the same temperature and pressure, contain an equal number of particles.

Alternatively, it can be stated that equal numbers of particles of all gases, when measured at the same temperature and pressure, occupy equal volumes.

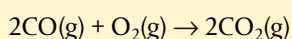
Using V for volume and n for number of moles:

$$V \propto n$$

This relationship enables us to relate gas volumes (of any gas) to the number of moles, and so to reacting ratios in equations.

Worked example

40 cm³ of carbon monoxide are reacted with 40 cm³ of oxygen in the reaction:



What volume of carbon dioxide is produced? (Assume all volumes are measured at the same temperature and pressure.)

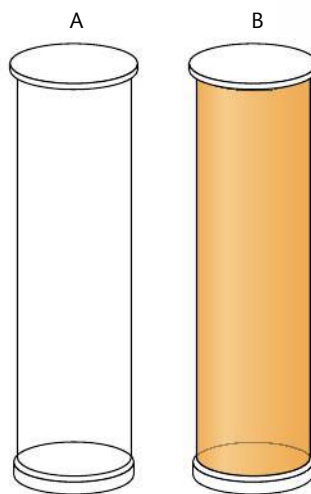


Figure 1.11 Flask A contains hydrogen molecules, flask B contains bromine molecules. The two flasks are under the same conditions of temperature and pressure.



Airbags have become a standard safety fitting in many vehicles. They are designed to act as a cushion or shock absorber by inflating rapidly on sudden impact of the vehicle during a collision. Airbags work on the principle of a chemical reaction triggered by the impact producing a gaseous product that causes a sudden volume change. The key reaction used is the conversion of sodium azide, NaN_3 , to nitrogen gas, N_2 . To avoid the production of dangerously reactive sodium metal, potassium nitrate, KNO_3 , and silicon dioxide, SiO_2 , are also included so that harmless silicates are produced instead.



Illustration of an airbag and seatbelt in action during a car accident. On impact, the airbag inflates and the seatbelt slows the forward force of the body, protecting the driver's head and chest.

Avogadro's law states that equal volumes of all gases at the same conditions of temperature and pressure contain equal numbers of particles:
 $V \propto n$.

CHALLENGE YOURSELF

- 5 Use the explanation on page 33 to deduce the chemical equations for the reactions taking place in a deployed airbag.

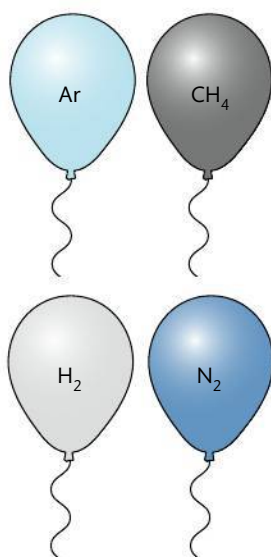


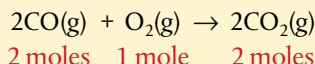
Figure 1.12 These four balloons are all filled with 1 dm³ of gas. At 25 °C and 100 kPa, they each contain 0.044 mol or 2.65×10^{22} atoms or molecules. Which balloon is the heaviest?

Note that STP (standard temperature and pressure) is not the same as the 'standard state', which is used in thermodynamic data and is explained in Chapter 5.

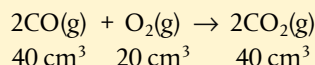
STP refers to a temperature of 273 K and a pressure of 100 kPa.

Solution

First identify the mole ratios in the equation:



The mole ratio is equal to the ratio of reacting gas volumes, so:



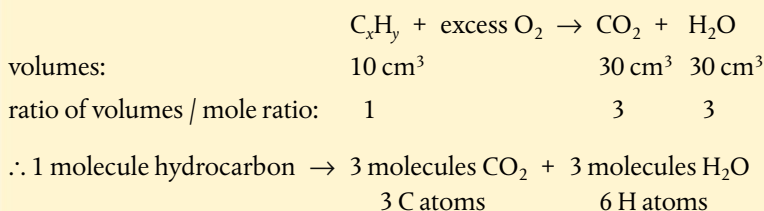
Therefore 40 cm³ of carbon dioxide are produced. (Oxygen is in excess by 20 cm³.)

Worked example

When 10 cm³ of a gaseous hydrocarbon (a compound containing only carbon and hydrogen) is burned in excess oxygen, the products consist of 30 cm³ of carbon dioxide and 30 cm³ of water vapour, measured under the same conditions of temperature and pressure. Determine the molecular formula of the hydrocarbon.

Solution

'Excess' oxygen indicates that the combustion reaction is complete.



The molecular formula is C₃H₆.

All gases under the same conditions have the same molar volume

On the basis of Avogadro's law, the volume occupied by one mole of any gas, known as the **molar volume**, must be the same for *all* gases when measured under the same conditions of temperature and pressure.

At standard temperature and pressure (STP), one mole of a gas has a volume of $2.27 \times 10^{-2} \text{ m}^3 \text{ mol}^{-1}$ ($=22.7 \text{ dm}^3 \text{ mol}^{-1}$). The conditions at STP are:

- a temperature of 0 °C (273 K)
- pressure of 100 kPa.

The molar volume can be used to calculate the amount of gas in a similar way to the use of molar mass earlier in this chapter. Here though the calculations are easier, as all gases have the same molar volume under the same conditions.

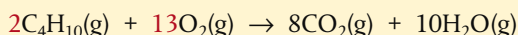
$$\text{number of moles of gas (n)} = \frac{\text{volume (V)}}{\text{molar volume}}$$

Worked example

What volume of oxygen at standard temperature and pressure would be needed to completely burn 1 mole of butane, C_4H_{10} ?

Solution

As always, start with the balanced equation and pick out the terms from the question.



mole / volume ratio: 1 6.5

6.5 moles of gas at STP have volume = $6.5 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 147.6 \text{ dm}^3$

Worked example

Calculate the volume occupied by 0.0200 g of He at standard temperature and pressure.

Solution

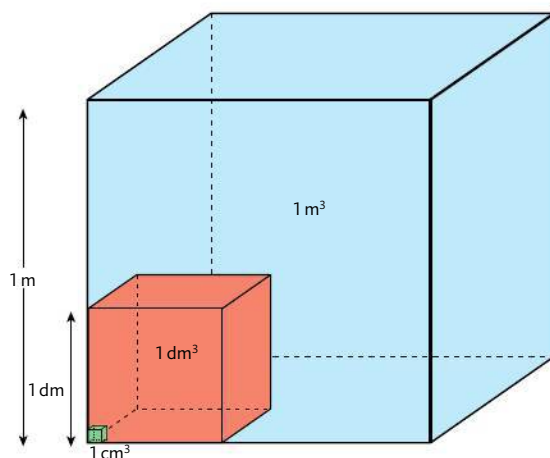
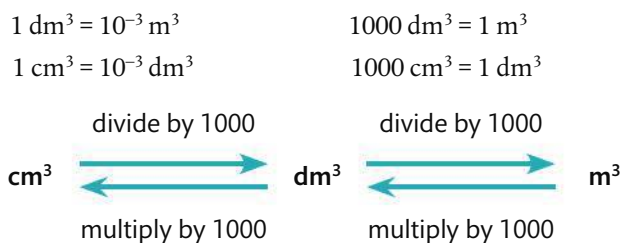
First convert the mass of He to moles.

$$n = \frac{m}{M} = \frac{0.0200 \text{ g}}{4.00 \text{ g mol}^{-1}} = 0.00500 \text{ mol}$$

$$\text{volume} = 0.00500 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 0.114 \text{ dm}^3$$

A note about units of volume

The metric unit m^3 is widely used in industrial and engineering calculations, but is too large to be convenient for many volume measurements in the laboratory. Instead, dm^3 and cm^3 are commonly used, so it is important to be able to interconvert these.



The litre (l or L) is widely used in place of dm^3 , and millilitre (ml or mL) in place of cm^3 . You will not be penalized for the use of these terms in an examination, but they will not be used in examination questions so it is essential that you know the correct use of m^3 , dm^3 , and cm^3 .



The value of the molar volume and the conditions for STP are given in section 2 of the IB data booklet.



CODATA (Committee on Data for Science and Technology) is an interdisciplinary scientific committee of the International Council of Science. It was established in 1966 to promote the worldwide compilation and sharing of reliable numerical data, such as the molar volume of a gas.

Figure 1.13 The three cubes are not to scale. $1 \text{ m}^3 = 1000 \text{ dm}^3 = 1\,000\,000 \text{ cm}^3$. In the laboratory, volumes are usually measured in cm^3 or dm^3 and often these measurements need to be converted to m^3 in calculations.

Exercises

45 How many moles are present in each of the following at STP?

(a) $54.5 \text{ dm}^3 \text{ CH}_4$

(b) $250.0 \text{ cm}^3 \text{ CO}$

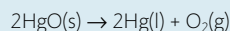
(c) $1.0 \text{ m}^3 \text{ O}_2$

46 What is the volume of each of the following at STP?

(a) 44.00 g N_2

(b) 0.25 mol NH_3

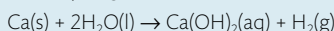
47 Pure oxygen gas was first prepared by heating mercury(II) oxide, HgO .



What volume of oxygen at STP is released by heating 12.45 g of HgO ?

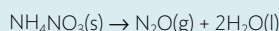
48 Which sample contains more molecules, 3.14 dm^3 of bromine, Br_2 , or 11.07 g of chlorine, Cl_2 when measured at the same temperature and pressure?

49 Calcium reacts with water to produce hydrogen.



Calculate the volume of gas at STP produced when 0.200 g of calcium reacts completely with water.

50 Dinitrogen oxide, N_2O , is a greenhouse gas produced from the decomposition of artificial nitrate fertilizers. Calculate the volume at STP of N_2O produced from 1.0 g of ammonium nitrate, when it reacts according to the equation:



NATURE OF SCIENCE

Early ideas on gas behaviour were suggested from the postulates of the kinetic theory, but could not advance without scientific evidence. This was provided by experimental work, mainly that of Boyle and Mariotte, Charles, and Gay-Lussac who contributed quantitative data based on testable predictions of how gases would respond to changes in temperature, volume, and pressure. In a fairly classic example of scientific process, the data supported the theory, and the theory explained the data. As a result there was wide acceptance of what became known as 'the gas laws' by the 18th century.

It is interesting to consider on the other hand why Avogadro's hypothesis was not widely accepted initially. Experiments led him to suggest that equal volumes of all gases at the same temperature and pressure contain the same number of molecules, but data to confirm this was somewhat lacking. In addition, his ideas conflicted with Dalton's atomic theory, which suggested that particles in gases could be only single atoms, not molecules as Avogadro proposed. It took the logical argument of Cannizzaro nearly 50 years later to show that Avogadro's hypothesis could be explained, and moreover used as a means to determine molecular mass. Following this, the relationship between gas volume and number of molecules became widely accepted and known as Avogadro's law. History has shown that the acceptance of scientific ideas by the scientific community is sometimes influenced by the time and manner of their presentation, as well as by their power to explain existing ideas.



The gas laws describe pressure, volume, and temperature relationships for all gases

The kinetic theory of matter, summarized on page 9, describes gases as largely empty space containing free moving particles of negligible volume having no inter-particle forces. This is often referred to as the **ideal gas** model. In effect it is an approximation, as no gas fits this description exactly, but it is nonetheless a useful means for predicting and interpreting the physical properties of gases under typical conditions of temperature and pressure. Later in this section we will explore situations when gas behaviour deviates from this ideal model.

You will be familiar with some behaviour of gases through everyday experiences such as blowing up a balloon or inflating a bicycle tyre to increase the pressure. Perhaps you have noticed how inflated balloons shrivel in colder conditions and expand when it is warmer? The interesting thing about these simple observations of the volume, pressure, and temperature of a gas is that they are not dependent on the chemical nature of the gas. In fact, all gases respond in the same way to changes in volume, pressure, and temperature when the mass of gas is fixed. These relationships are summarized as the **gas laws** and are discussed below.

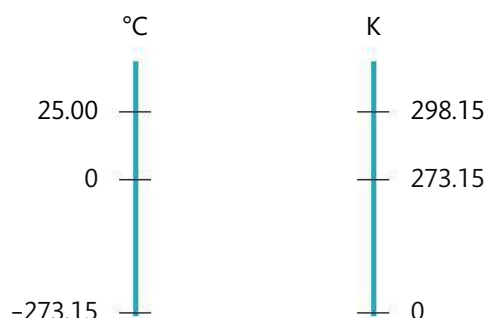
An ideal gas is one that obeys the ideal gas laws.



A note about units of temperature

In all work on gases, it is essential to use values for temperature recorded in Kelvin (K), not in Celsius (°C). Temperature in Kelvin is known as the **absolute temperature**, and is based on a scale where absolute zero, 0 K, is the point of zero kinetic energy of particles. This coincides with $-273.15\text{ }^{\circ}\text{C}$. As the interval on the Kelvin scale is the same as that on the Celsius scale, conversion between the two simply involves addition or subtraction of 273.15 (commonly approximated to 273).

$$\text{temperature (K)} = \text{temperature (}^{\circ}\text{C)} + 273.15$$



William Thomson (1824–1907), who became known as Lord Kelvin later in life, completed most of his work at the University of Glasgow, Scotland. His concept of the absolute temperature scale followed from his recognition of the relationship between heat energy and the ability to do work. The existence of a minimum possible temperature at which no heat can be extracted from the system and so no work done, led him to the definition of absolute zero in 1848. This in turn led to the formulation of the laws of thermodynamics. Kelvin is considered one of the great scientists of the 19th century, and is buried next to Isaac Newton in London.

A note about units of pressure

The SI unit of pressure is the Pascal (Pa), which is equal to N m^{-2} . Pressure is now commonly given in bars, where $10^5\text{ Pa} = 1\text{ bar}$, as this is conveniently close to 1 atmospheric pressure.

NATURE OF SCIENCE

The definition of absolute zero (0 K) as the temperature where a substance has no kinetic energy, suggests it is the point at which all motion in particles ceases. It is the lowest possible temperature. Zero Kelvin has not been achieved, although modern technologies, which improve cooling methods and use magnets to contain the gas, are helping scientists to reach values ever closer to this. Researchers in Finland have achieved temperatures as low as 100 pK ($1 \times 10^{-10}\text{ K}$) in a piece of rhodium metal. Ultra-low temperature research has led to observations of phenomena such as quantum fluid behaviour and superconductivity, and could lead to improvements in precision measurements such as those used in atomic clocks and sensors for gravity and rotation. Science progresses as improvements in technology give access to new information, and studies in one field open up possibilities in another. Superconductivity is discussed further in Chapter 12.

1 Relationship between volume and pressure

The volume of a gas is always the volume of its container as the particles spread out fully. Its pressure is the result of the particles colliding with the walls of the container, and will increase when the frequency or energy of these collisions increases.

If the temperature is held constant, it is found that increasing the pressure on a fixed mass of gas decreases its volume. In other words, the pressure of a gas is inversely proportional to its volume, and the product of pressure and volume is a constant.

TOK

The ideal gas equation can be deduced from a small number of assumptions of ideal behaviour. What is the role of reason, perception, intuition, and imagination in the development of scientific models?



▲ The so-called mentos-soda fountain reaction. Dissolved carbon dioxide in the soda changes quickly into gas in the presence of the candy mint mentos. The sudden pressure change causes the soda to be ejected in a fountain of foam.



Different countries continue to use a variety of units for pressure, including millimetres of mercury (mm Hg), torr, pounds per square inch (psi), and atmosphere (atm).



In all the examples here, use of the term 'pressure' assumes 'absolute pressure', which is referenced against a perfect vacuum. This is in contrast to the gauge pressure, which is measured relative to atmospheric pressure.

$$P \propto \frac{1}{V}$$

$$PV = \text{a constant}$$

Figure 1.14 As the pressure on a gas is increased, its volume decreases proportionately.

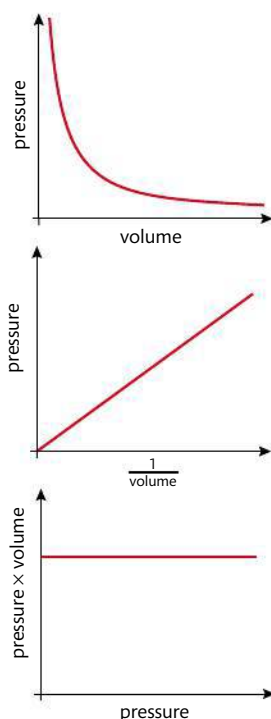
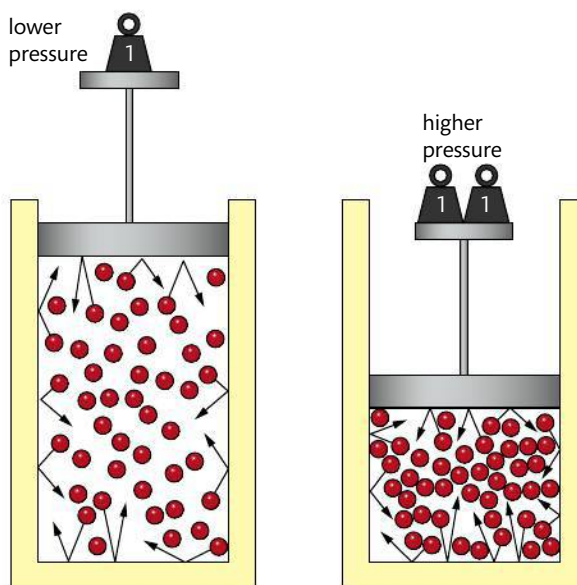


Figure 1.15 Boyle's law. Gas pressure is inversely proportional to its volume.

Figure 1.16 Charles' law. Gas volume is proportional to the absolute temperature. Note the dotted line represents an extrapolation, as data at temperatures down to 0 K are not obtainable.

$$V \propto T$$

$$\frac{V}{T} = \text{a constant}$$



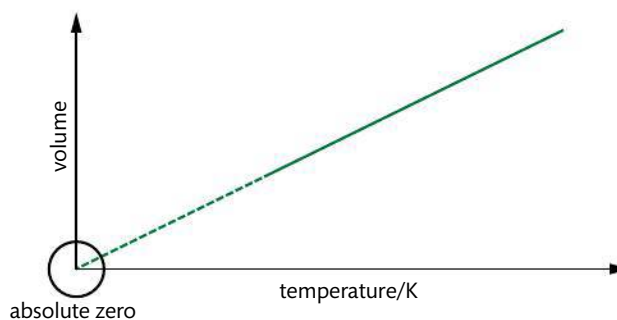
$$P \propto \frac{1}{V}$$

$$PV = \text{a constant}$$

This relationship is often known as **Boyle's law**, as it was first established by Robert Boyle in 1662. Application of this relationship is found in the compression of gases under pressure, often useful in transport and storage.

2 Relationship between volume and temperature

An increase in temperature represents an increase in the average kinetic energy of the particles. If the pressure is held constant, it is found that increasing the temperature of a fixed mass of gas increases its volume. In other words, the volume of a gas is directly proportional to its absolute temperature, and volume divided by absolute temperature is a constant.



$$V \propto T$$

$$\frac{V}{T} = \text{a constant}$$

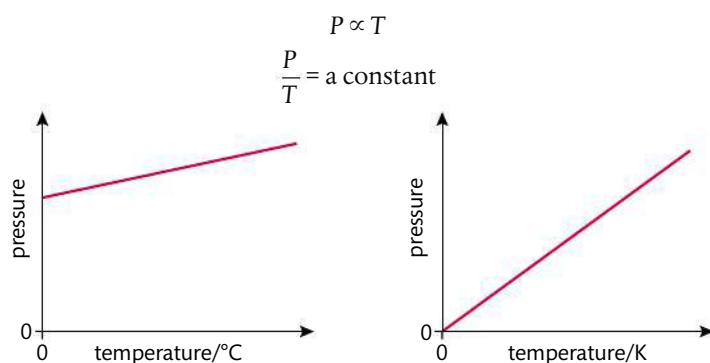
TOK

The SI base unit of temperature is the Kelvin, which has a natural basis for its definition of zero. By contrast, zero in the imperial scale of degree Celsius is arbitrarily defined. What are the implications of using a scale based on natural or arbitrary values?

This relationship is often known as **Charles' law**, as it was first established by Frenchman Jacques Charles in the late 18th century. You can demonstrate this relationship by immersing dented table tennis balls in warm water. As the air inside the ball equilibrates to the temperature of the water, it expands, pushing the dents out on the surface.

3 Relationship between pressure and temperature

An increase in temperature increases the average kinetic energy of the particles. The particles move faster and collide with the walls of the container with more energy and more frequency, raising the pressure. If the volume is held constant, it is found that increasing the temperature of a fixed mass of gas proportionately raises its pressure. In other words, the pressure of a gas is directly proportional to the absolute temperature, and pressure divided by temperature is a constant.



NATURE OF SCIENCE

Scientists work to contribute to a common body of knowledge. Results are shared through publication, considered, and used as the basis for further studies by other scientists. Although many scientific discoveries, such as the gas laws, are named after the key scientist involved, in reality the theories and laws of science stand apart from the individual discoverers. Without Charles and Boyle, the relationships between pressure, volume, and temperature of a gas would still exist. This is very different from the arts. Without Shakespeare, there would have been no *Hamlet*, without Picasso, no *Guernica*.



Balloons dipped in a mixture of dry ice (solid CO_2) and water rapidly deflate as the air inside contracts at low temperature.

Figure 1.17 Gas pressure is proportional to the absolute temperature.



Always be certain to use temperature in Kelvin only when applying these relationships.

The label on the aerosol can warns of the dangers of exposing the pressurized contents to high temperature.



$$P \propto T$$

$$\frac{P}{T} = \text{a constant}$$



Pressurized cans, such as soda or beer, often carry a warning to be stored in a cool place. This is because at higher temperatures the pressure inside the can at fixed volume is able to rise to the point of causing the can to explode.

These three gas laws applied to a fixed mass of gas can be summarized as follows:

$$P \propto \frac{1}{V} \text{ at constant temperature}$$

$$V \propto T \text{ at constant pressure}$$

$$P \propto T \text{ at constant volume}$$

These can be combined to give one equation for a fixed mass of gas:

$$\frac{PV}{T} = \text{a constant or}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \text{ (where 1 and 2 refer to initial and final conditions respectively)}$$

Application of this enables gas volume, pressure, and temperature to be calculated as conditions change.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$



Worked example

What happens to the volume of a fixed mass of gas when its pressure and its absolute temperature are both doubled?

Solution

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$P_2 = 2 \times P_1$ and $T_2 = 2 \times T_1$, so these can be substituted into the equation:

$$\frac{P_1 V_1}{T_1} = \frac{2P_1 V_2}{2T_1}$$

We can cancel P_1 and T_1 from both sides and 2s on the right side, leaving

$$V_1 = V_2$$

The volume does not change.

Worked example

The molar volume of a gas at STP is $22.7 \text{ dm}^3 \text{ mol}^{-1}$. Calculate the molar volume at 25°C at the same pressure.

Solution

As the pressure is not changing, we do not need to insert P_1 and P_2 into the combined gas equation. Temperature must be converted from $^\circ \text{C}$ to K.

$$T_1 = 273 \text{ K}, T_2 = 25 + 273 = 298 \text{ K}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{22.7 \text{ dm}^3}{273 \text{ K}} = \frac{V_2}{298 \text{ K}}$$

$$V_2 = \frac{298 \times 22.7 \text{ dm}^3}{273} = 24.8 \text{ dm}^3$$

Note that temperature must be in Kelvin in these calculations. However, the units of volume and pressure are not specified, so long as they are consistent on both sides of the equation.



The ideal gas equation is derived from the combined gas equation and Avogadro's law

The combined gas equation tells us that

$$\frac{PV}{T} = \text{a constant}$$

The value of the constant is directly proportional to the fixed mass of gas, or the number of moles, n .

$$\text{So } \frac{PV}{T} \propto n$$

This can be made into an equation by introducing a constant, **R**, known as the **universal gas constant**.

$$\therefore \frac{PV}{T} = nR, \text{ which is usually written as } PV = nRT$$

This equation is known as the **ideal gas equation**, and is given in Section 1 of the IB data booklet. The value of R can be calculated by substituting known values into the equation, such as those for the molar volume of a gas at STP. In this case:

$$P = 10^5 \text{ Pa (N m}^{-2}\text{)}, V = 2.27 \times 10^{-2} \text{ m}^3, T = 273 \text{ K}, n = 1$$

$$\therefore 10^5 \text{ N m}^{-2} \times 2.27 \times 10^{-2} \text{ m}^3 = 1 \text{ mol} \times R \times 273 \text{ K}$$

$$R = 8.31 \text{ N m K}^{-1} \text{ mol}^{-1} \text{ or } 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

This value for R , the gas constant, is given in Section 2 in the IB data booklet and should be used for all calculations involving the ideal gas equation.

Use of the ideal gas equation enables us to calculate how systems respond to changes in pressure, volume, and temperature, and to calculate molar mass. Gas density can also

be derived by applying the relationship $\text{density} = \frac{\text{mass}}{\text{volume}}$.

These calculations usually involve simply substituting values into the equation, but the use of units needs special attention here. The guidelines below, based on the use of SI units only, should help you avoid some of the common mistakes that arise.

- Pressure, P : must be in Pa (N m^{-2}); if kPa are given, multiply by 10^3 .
- Volume, V : must be in m^3 ; if dm^3 are given, divide by 10^3 , if cm^3 are given divide by 10^6 .
- Number of moles, n : this is often derived by application of $n = \frac{m}{M}$.
- Temperature, T : must be in Kelvin; if $^{\circ}\text{C}$ is given, add 273.15.

Worked example

A helium party balloon has a volume of 18.0 dm^3 . At 25°C the internal pressure is 108 kPa . Calculate the mass of helium in the balloon.

Solution

First ensure all data are in SI units:

$$P = 108 \text{ kPa} = 108 \times 10^3 \text{ Pa}$$

$$V = 18.0 \text{ dm}^3 = 18.0 \times 10^{-3} \text{ m}^3$$

$$T = 25^{\circ}\text{C} = 298 \text{ K}$$



The ideal gas equation is
 $PV = nRT$



Many sources give data for ideal gas law questions in non-SI units such as atmosphere, which require a mathematical conversion. You will not be expected to be familiar with these conversions in IB examinations.

As this example shows, if pressure is given in kPa and volume is given in dm^3 , the same answer is obtained if these values are used directly in the ideal gas equation in place of Pa and m^3 .



$$PV = nRT$$

$$108 \times 10^3 \text{ Pa} \times 18.0 \times 10^{-3} \text{ m}^3 = n \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1} \times 298 \text{ K}$$

$$\therefore n(\text{He}) = 0.785 \text{ mol}$$

$$\therefore \text{mass}(\text{He}) = nM = 0.785 \text{ mol} \times 4.00 \text{ g mol}^{-1} = 3.14 \text{ g}$$

Worked example

A sample of gas has a volume of 445 cm^3 and a mass of 1.500 g at a pressure of 95 kPa and a temperature of 28°C . Calculate its molar mass.

Solution

Substitute $n = \frac{m}{M}$ into the ideal gas equation, and rearrange to solve for M .

$$M = \frac{mRT}{PV}$$

Ensure all data are in SI units:

$$P = 95 \text{ kPa} = 95 \times 10^3 \text{ Pa}$$

$$V = 445 \text{ cm}^3 = 445 \times 10^{-6} \text{ m}^3$$

$$T = 28^\circ\text{C} = 301 \text{ K}$$

$$\therefore M = \frac{1.500 \text{ g} \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1} \times 301 \text{ K}}{95 \times 10^3 \text{ Pa} \times 445 \times 10^{-6} \text{ m}^3} = 88.8 \text{ g mol}^{-1}$$

Worked example

A gas has a density of 1.65 g dm^{-3} at 27°C and 92.0 kPa . Determine its molar mass.

Solution

density data $\Rightarrow 1.65 \text{ g}$ occupies 1.00 dm^3

As in the example above, substitute data with correct units into $M = \frac{mRT}{PV}$

$$\therefore \frac{1.65 \text{ g} \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}}{92.0 \times 10^3 \text{ Pa} \times 1.00 \times 10^{-3} \text{ m}^3} = 44.7 \text{ g mol}^{-1}$$

Blowing up a balloon increases its volume as the number of particles increases. What do you think would happen to this inflated balloon on the top of a very high mountain?



Experiment to calculate the molar mass of carbon dioxide by application of the ideal gas equation

Full details with a worksheet are available online.

A known mass of $\text{CuCO}_3(\text{s})$ is heated and the gas evolved collected by displacement of water. The volume of the gas, the room temperature, and pressure are recorded.

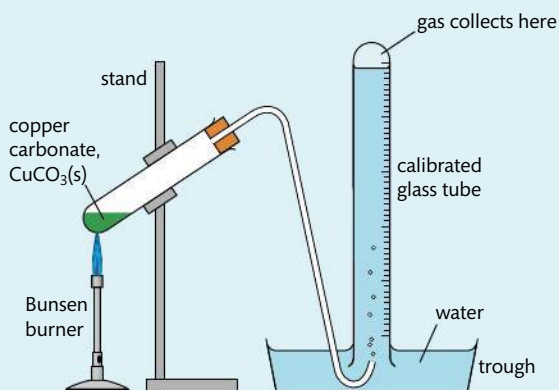
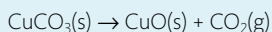


Figure 1.18 Experiment to calculate the molar mass of carbon dioxide.

Sample results

	Trial 1	Observations
mass of boiling tube + CuCO_3 before heating / $\text{g} \pm 0.001$	33.910	CuCO_3 is a green powder
mass of boiling tube + CuCO_3 after heating / $\text{g} \pm 0.001$	33.822	the contents of the tube are black after heating
mass change (mass CO_2) / $\text{g} \pm 0.002$	0.088	
volume of gas collected / $\text{cm}^3 \pm 0.1$	38.1	the gas collected is colourless
temperature / $\text{K} \pm 0.1$	293.0	
pressure / $\text{Pa} \pm 0.1$	101.3 kPa	

Processed data

$$M(\text{CO}_2) = \frac{mRT}{PV} = \frac{0.088 \text{ g} \times 8.31 \text{ J mol}^{-1} \text{ K}^{-1} \times 293.0 \text{ K}}{101.3 \text{ kPa} \times 0.0381 \text{ dm}^3}$$

experimental value $M(\text{CO}_2) = 55.5 \text{ g mol}^{-1}$

theoretical value $M(\text{CO}_2) = 44.01 \text{ g mol}^{-1}$

$\therefore$ % error = 26.1%

The percentage error can be analysed in consideration of systematic errors such as:

- gas collected may not be pure CO_2 ;
- CO_2 may be soluble in water;
- air in the tube is collected with the gas;
- gas collected has not equilibrated to room temperature.

Modifications to the experimental design should suggest ways to reduce the impact of these errors. Note that repeat trials and error propagation are not shown here.



NATURE OF SCIENCE

Scientists often use models, which may be simple or complex, to explain concepts that are not observable. By their nature, models are thinking tools and so they all have limitations. Models are often tested against experimental results, and should be able to explain phenomena that are seemingly different from the ones used to develop the model. Sometimes, when a model does not seem consistent with data or observations, it may need to be modified or replaced. At other times it can be retained, with limitations to its usefulness agreed.

The ideal gas model has all of these characteristics. While it provides a useful conceptual image of gas behaviour, and is consistent with some testable predictions, there are limitations to this, as we will see below. These do not mean, however, that the model should be discarded – rather they alert us to the fact that the model is in some ways an over-simplification, and cannot apply accurately under all conditions.

Exercises

- 51 A 2.50 dm³ container of helium at a pressure of 85 kPa was heated from 25 °C to 75 °C. The volume of the container expanded to 2.75 dm³. What was the final pressure of the helium?
- 52 After a sample of nitrogen with a volume of 675 cm³ and a pressure of 1.00×10^5 Pa was compressed to a volume of 350 cm³ and a pressure of 2.00×10^5 Pa, its temperature was 27.0 °C. Determine its initial temperature.
- 53 The absolute temperature of 4.0 dm³ of hydrogen gas is increased by a factor of three and the pressure is increased by a factor of four. Deduce the final volume of the gas.
- 54 To find the volume of a flask, it was first evacuated so that it contained no gas at all. When 4.40 g of carbon dioxide was introduced, it exerted a pressure of 90 kPa at 27 °C. Determine the volume of the flask.
- 55 An unknown noble gas has a density of 5.84 g dm⁻³ at STP. Calculate its molar mass, and so identify the gas.
- 56 A 12.1 mg sample of a gas has a volume of 255 cm³ at a temperature of 25.0 °C and a pressure of 1300 Pa. Determine its molar mass.
- 57 Which has the greater density at STP, hydrogen or helium?
- 58 Calculate the volume of oxygen at STP required for the complete combustion of 125 g of octane, C₈H₁₈, to form carbon dioxide and water.
- 59 A sample of an unknown gas with a mass of 3.620 g occupied a volume of 1120 cm³ at a pressure of 99 kPa and a temperature of 25.0 °C. The sample contained 2.172 g O and 1.448 g S. Determine the empirical and molecular formula of the gas.
- 60 A road cyclist pumps his tyres up very hard before a trip over a high mountain pass at high altitude. Near the summit one of his tyres explodes. Suggest why this may have occurred.

Real gases show deviation from ideal behaviour

An ideal gas is defined as one that obeys the ideal gas law $PV = nRT$ under all conditions. This means that for one mole of gas, the relationship PV/RT should be equal to 1. So a graph of PV/RT against P for one mole of an ideal gas is a horizontal line of intercept 1 (Figure 1.19).

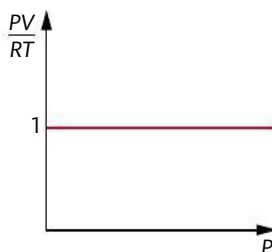


Figure 1.19 For one mole of an ideal gas, the relationship PV/RT is a constant at all pressures.

But, as we noted earlier, there is no such thing as an ideal gas. All gases, known as **real gases**, deviate to some extent from ideal behaviour. So, for real gases the value of PV/RT for one mole will vary. An example of the extent of this variation from 1 at different conditions is shown in Figure 1.20.